C-HI Two-node Restart Rehearsal Quick-look

Date: 2026-05-09

Summary

This report records the first production-template, multi-node restart rehearsal for the particles-enabled `C-HI` Stewart-cluster run on `kunanyi`.

Run facts:

- source restart job: `690484.kunanyi-ohpc`
- job name: `c_hi_2node_env`
- scheduler result: `Exit_status = 0`
- resources: `select=2:ncpus=28:mpiprocs=28`
- nodes: `n038` and `n083`
- MPI ranks: `56`
- PBS walltime used: `01:45:08`
- PBS CPU time used: `45:43:18`
- PBS peak memory: `18587788kb`
- PBS virtual memory: `27291908kb`
- template restart mode: `RUN_KIND=h5_restart_2`
- PLUTO command: `-i pluto.ini -e environment.ini -maxtime 1.5 -h5restart 2`

The job restarted from `data.0002.dbl.h5` and advanced the run to a new restart-capable output, `data.0003.dbl.h5`, at `2.1501 Myr`. The run wrote a fresh `markers/DONE` file and no `FAILED` marker was present after completion.

This is still an early run-health diagnostic rather than a science result, but it is an important gate: the current production template has now run the particles-enabled `C-HI` case across two `kunanyi` nodes, with HDF5 restart and HDF5 output intact.

Launch Fixes Captured

Two intentionally small failed attempts preceded the successful job:

- `690481.kunanyi-ohpc` showed that OpenMPI did not infer the requested `56` PBS slots from the scheduler allocation.
- `690483.kunanyi-ohpc` showed that the second node did not inherit the HDF5 runtime library path, failing on `libhdf5.so.310`.

The production template now writes an explicit OpenMPI hostfile from `PBS_NODEFILE` and exports `PATH`, `LD_LIBRARY_PATH`, and HDF5-related environment variables through `mpirun`. The successful `690484` hostfile was:

```
n038.kunanyi slots=28
n083.kunanyi slots=28
```

This means the multi-node launch issue has been reduced to a documented template fix, not a standing blocker.

Output Catalogue

The current HDF5 catalogue now contains two regular science-cadence float outputs, plus three walltime-forced double restart outputs.

Output	Type	Time (Myr)	Step	Role
0	`dbl.h5`	0.0000	0	initial restart-capable output
1	`dbl.h5`	0.4474	583	first walltime-forced restart output
2	`dbl.h5`	1.0102	1146	second walltime-forced restart output
3	`dbl.h5`	2.1501	2267	two-node walltime-forced restart output
0	`flt.h5`	0.0000	0	initial float output
1	`flt.h5`	0.4474	583	first walltime-forced float output
2	`flt.h5`	0.9990	1135	nominal 1 Myr science-cadence output

Output	Type	Time (Myr)	Step	Role
3	`flt.h5`	1.0102	1146	second walltime-forced float output
4	`flt.h5`	1.9996	2120	nominal 2 Myr science-cadence output
5	`flt.h5`	2.1501	2267	two-node walltime-forced float output

The corresponding CSV is stored at `docs/science/assets/c_hi_2node_quicklook_2026-05-09/output_catalogue.csv`.

Particle outputs now extend through `particles.0218.dbl`.

Quick-look Extraction

A one-core compute-node extraction job inspected the nominal 2 Myr float output:

- quick-look extraction job: `690559.kunanyi-ohpc`
- inspected file: `/scratch/bwbrook/xray_cavities_runs/2026-05-08-c-hi-segment/C-HI/particles/problem/outputs/data.0004.flt.h5`
- timestep: `Timestep_4`
- PLUTO code time: `613.07384055587761`
- physical time: `1.9996 Myr`
- extraction result: `Exit_status = 0`

The compact evidence bundle was retrieved to:

```
build/science/kunanyi_stage_2026-05-08/c_hi_2node_690484_quicklook_690559_compact_2026-05-09
```

SHA-256:

```
e002e0df31f5f2749ce823f7ca5b897c4b6b4cb7b322be9e29e9fb8cf9b045bb
```

Tracer Extent

At the default Level 2 truth threshold, `tracer > 5e-3`, the tracer-defined region has grown smoothly from the previous early outputs:

Snapshot	Time (Myr)	Default-threshold x/y extent	Default-threshold z extent
first segment quick-look	0.4474	`+/-2.33 kpc`	`+/-6.22 kpc`
first restart quick-look	0.9990	`+/-3.84 kpc`	`+/-10.1 kpc`
two-node quick-look	1.9996	`+/-5.63 kpc`	`+/-16.5 kpc`

Detailed 2 Myr thresholds:

Threshold	Cells	x range (kpc)	y range (kpc)	z range (kpc)
`1e-3`	6351456	-5.82 to 5.82	-5.82 to 5.82	-16.5 to 16.5
`5e-3`	5908752	-5.63 to 5.63	-5.63 to 5.63	-16.5 to 16.5
`1e-2`	5697720	-5.53 to 5.53	-5.53 to 5.53	-16.1 to 16.1

Figures

Full-domain mid-plane maps:

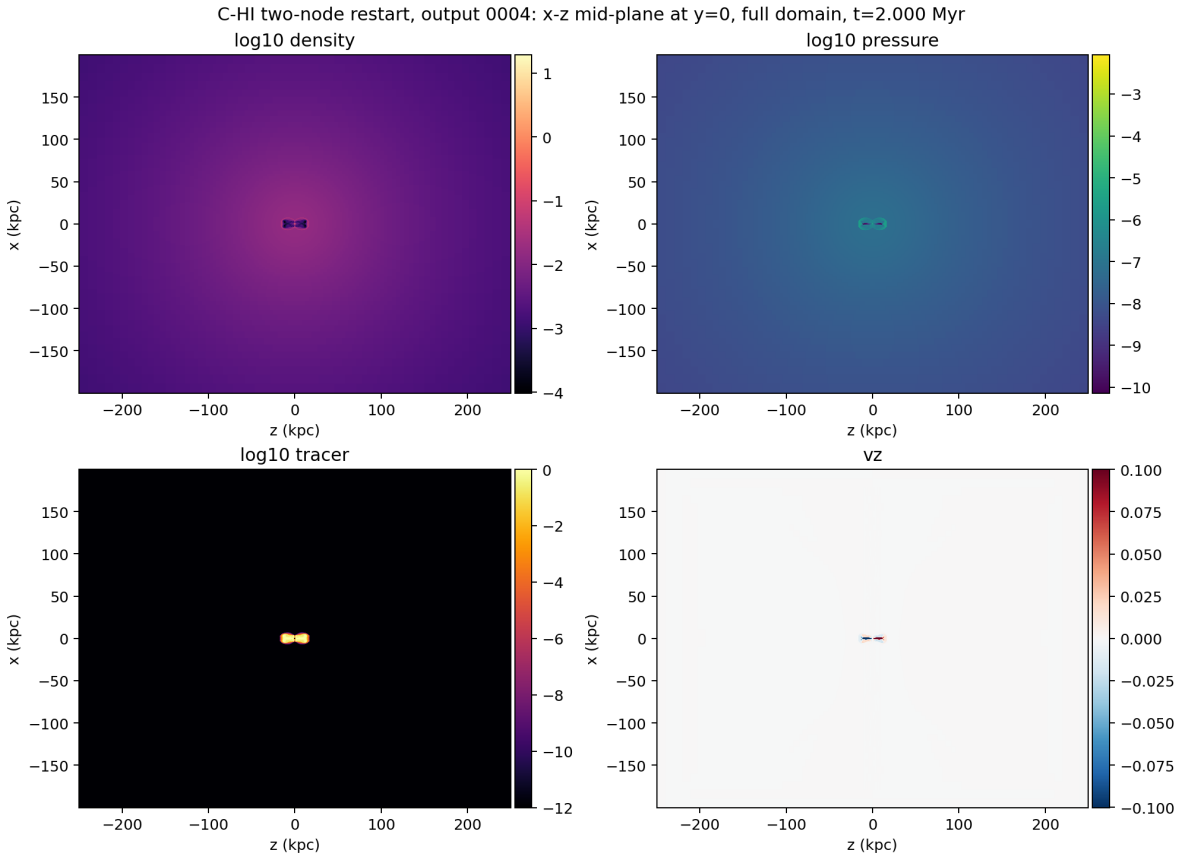


Figure 1: C-HI two-node full-domain quick-look maps

Central zoom:

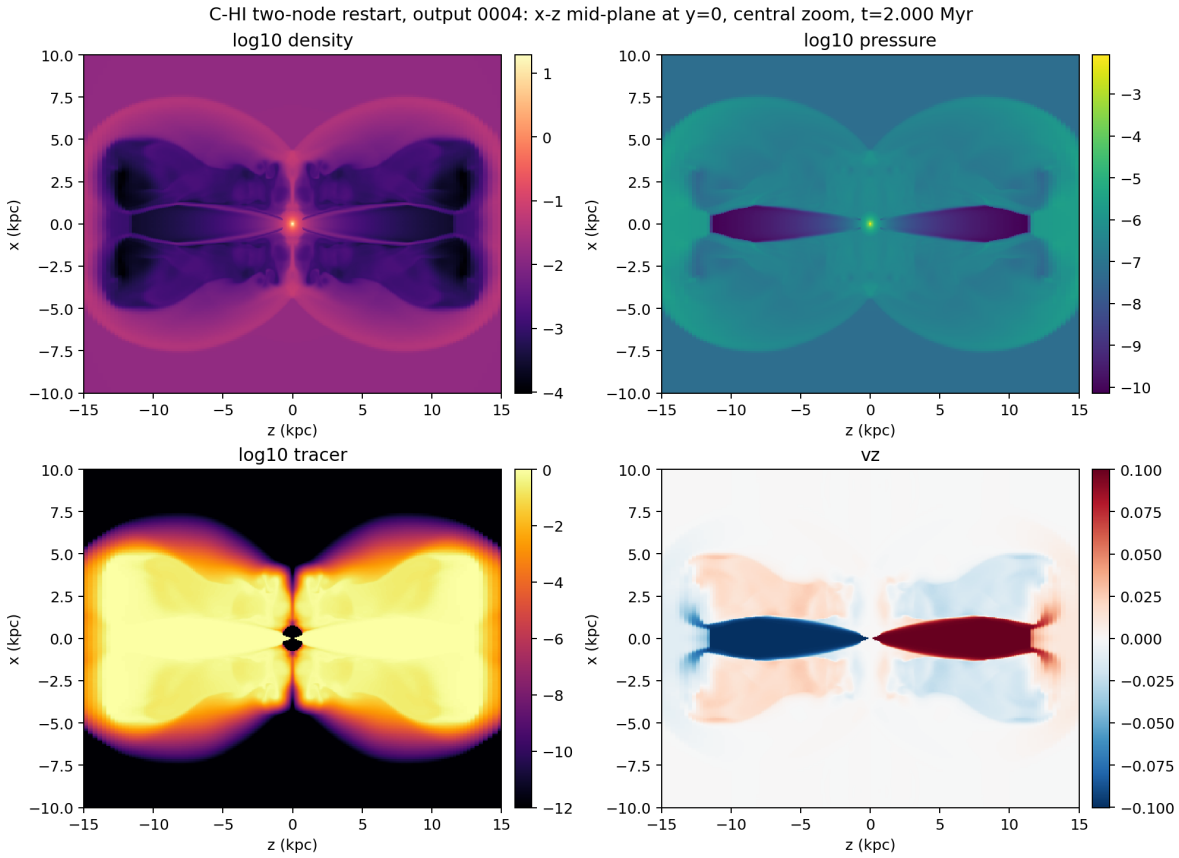


Figure 2: C-HI two-node central quick-look maps

Jet-axis profiles:

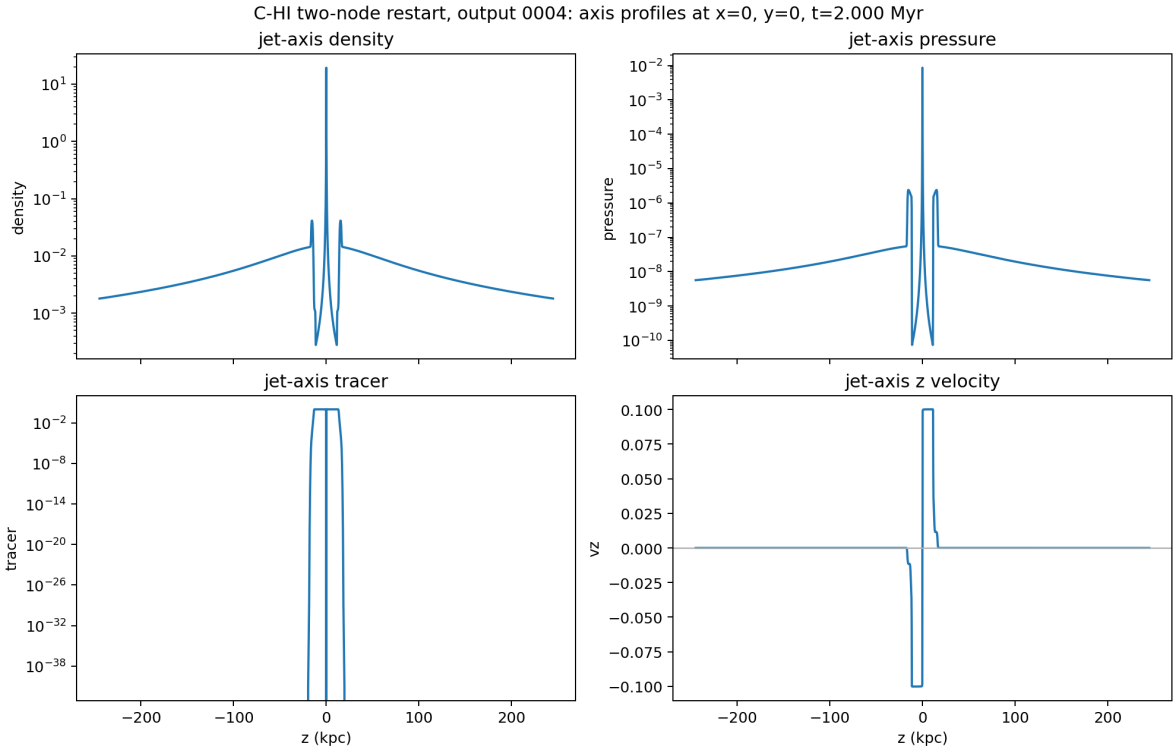


Figure 3: C-HI two-node quick-look axis profiles

Interpretation

The two-node rehearsal passes the intermediate `kunanyi` gate:

- HDF5 restart from `data.0002.dbl.h5` worked via `-h5restart 2`.
- OpenMPI launched across two PBS nodes using a generated hostfile.
- The HDF5 runtime environment propagated to the second node.
- PLUTO stopped through the requested `-maxtime 1.5` path.
- A new restart-capable `dbl.h5` output was written.
- The nominal 2 Myr `flt.h5` output was written and extracted successfully.
- Tracer, density, pressure, and `v_z` quick-looks remain symmetric and far from the outer grid boundaries.

The next compute rung should be a longer early-snapshot continuation, not yet a full 48-hour production push. The near-term target is to reach the first `3-5` regular science snapshots cleanly, then assemble the first aligned truth/X-ray/radio smoke package around a declared snapshot and view.